Welcome to OpenDev Etherpad!

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ai + ops

- copilot :
 - summrize , generate scripts : tf scripts , ansible playbooks , kubernets , manifest, cicd scripts..genrate commands
 - troubleshooting , errors : reduced our time in finding out most probable cause
 - explaination

monitoring tool : hundreds and hundrds of alerts :
 duplication
 not important
 critical
 fake

 app stops , server crash : 50% cpu
 metrics , logs---------- hp monitoring system detect an anomaly drift ---------- manually alerts (ai , machine learning (classify/predictions)) ---------- creating a ticket (priority) -- automated workflow (cicd) restart
 -- assigning to a prodn engg

cpu : hundreds of alerts
noise : fake alerts , reducdant alert, duplicate

alert fatigue !! : make errors

ai : thiusands of alerts: tell the engg : this can be most

thousands lines log of m y app:

toil ? repetive work

predictive analysis :
aggregation :
correlation

probable root cause : direction in which ivestigate

github , bitbucket , gitlab : version control systems ...

aiops ,, why in operations ?

git ..track of changes

java_01.zip
java_v2.zip
java_v3.zip

20 backups

git was introduced , track the changes , vcs

git -h
 499 git brranch
 500 git brranches
 501 git branches
 502 git branch
 503 git fetch origin
 504 git branch -r
 505 git branch
 506 git switch --track origin/feature/login
 507 git switch --track origin/feature/improve-cpu-runbook
 508 git branch
 509 ls
 510 cd runbooks/
 511 ls
 512 cat high-cpu-utilization.md
 513 git checkout main
 514 git branch
 515 ls
 516 cat high-cpu-utilization.md
 517 ls
 518 git checkout feature/improve-cpu-runbook
 519 vi high-cpu-utilization.md
 520 git add .
 521 git commit -m " removed the last line"
 522 git push origin feature/improve-cpu-runbook
 523 git log
 524 clear

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investigation : runbooks ..., tf scripts, kubernetes yaml, linux shells scripts.... reliable place ; version control

application , vms , netwrk devices , kubernetes , containers [raw DATA] metrics , logs , traces] ---------- data managemnt preprocessing ---------- data lake (git lfs , s3 , azure blob storage)---------- MODEL (train accordingly), predict /classify/ dynatrace api

mngemnt : cleaning / preprocessing ..
missing values
different formats

data :
metric : 90% .....
log : brief description ; errors , warning , debug
trace : api ---- frontend -----backend ------ s3 ------ image ---- frontend ---------raman

clean data ----------train the model with thousands varied alerts---------- aiops : machine learning/ ai llms

predictive analylises
anaomoly detection
aggregation

raw - ingest-------- s3 datalake ---------- preprocessing app (python code /cicd pipeline) ----------

saving ur booking in a queue (500/1000)--------queeue ---------- loads (server 90 %)

generated raw data synthectically ---------- inspect that data----------validation stage & preprocessing data which has been generated [pandas librabry]

E : EXTRACTION FROM PRODN SYSTEMS
L : load/transfer it to s3 data lake
T : trasnforming/preprocessing data

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AI : medical , IT , robotics , etc

trad. machine learning (predict /classification) [statistics , ml algorthms : knn , svm randon forest , logictic regression , etc]

- Deep learning (neural network) pattern relationships
Gen AI (predict /classify / generate text as well)

COMPUTE !! gpu s : ai datacenter : train thier own ai models

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download that zip ; extract it

open that folder inside ur vs code :

502 python -m venv .venv
503 source .venv/Scripts/activate
504 pip list
505 pip install -r requirements.txt

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sklearn library : islotaion forest

train the machine with isolation forest (clean dataset 20) : serving model [learn those statiscal patterns] .... gpu

test .....serving model [learned those statiscal patterns] inference to new data ....

cold storage ~ glacier --- cheapest
standard : hot standard.....most costliest

summarize something
review
refactoring scripts
operations scripts

chatgpt , kiro , microsoft copilot , github copilot , : applications ....

llms/models : foundation models :

open ai : gpt 5.6 luna ; parameters :
anthropic : haiku , fable , etc
deepseek : qwen
huggingface : ...
aws : titan
microsoft : llama
google : gemini, argon

reatreval : chatbot , it wl forward the request to vector database
augmnetation : thinking
genration

s3 bucket

console access ...
aws sdk botocore
aws cli ...

terraform script (iac)
cloudformation .....native to aws
bicep..azure

s3 bucket , sg , machines, eks cluster .....30 resources ..

code ...(tf script) . trackable ..github..

reusable ..whenever to create that infra